

CS & IT ENGINEERING



Computer Network

Introduction

Lecture No. - 05



By - Abhishek Sir





Recap of Previous Lecture



Topic

Transport Layer

Topic

Network Layer

Topic

Data Link Layer



Topics to be Covered



Topic

Data Link Layer

Topic

Physical Layer

Topic

Network Topology

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : Data Link Layer



Data Link layer services :-

1. Framing ✓

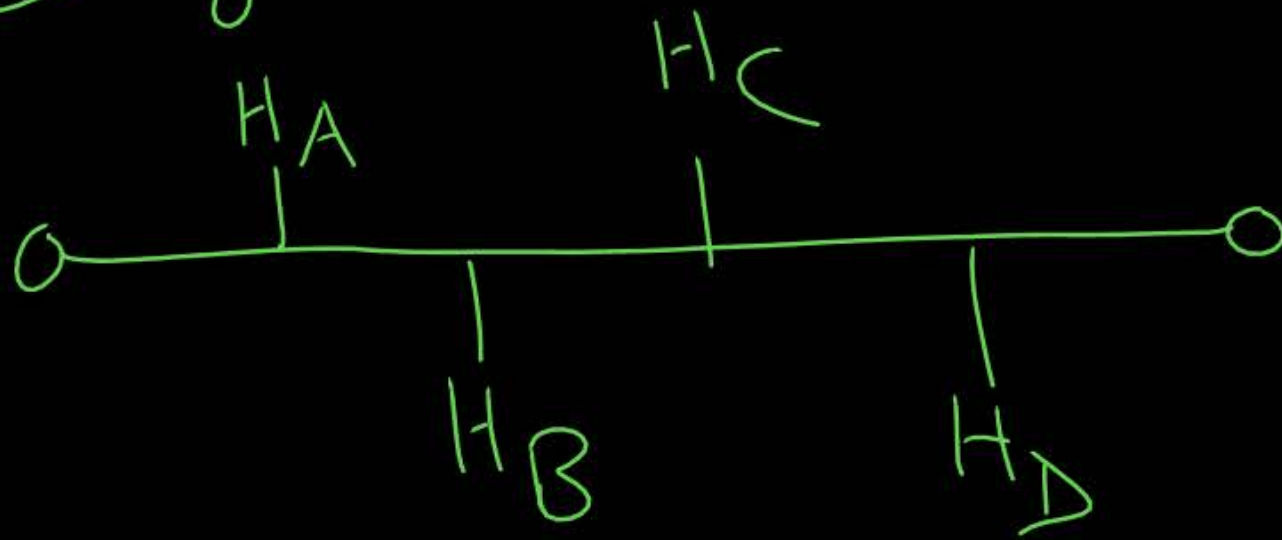
2. Error Control

3. Flow Control

4. [Access Control]

→ point to point connection

[channel sharing]





Topic : Data Link Layer



Data Link Layer PDU : **“Frame”** ✓

→ Sender : Encapsulate datagram into frame, ✓ →
Frames passes to physical layer for transmission

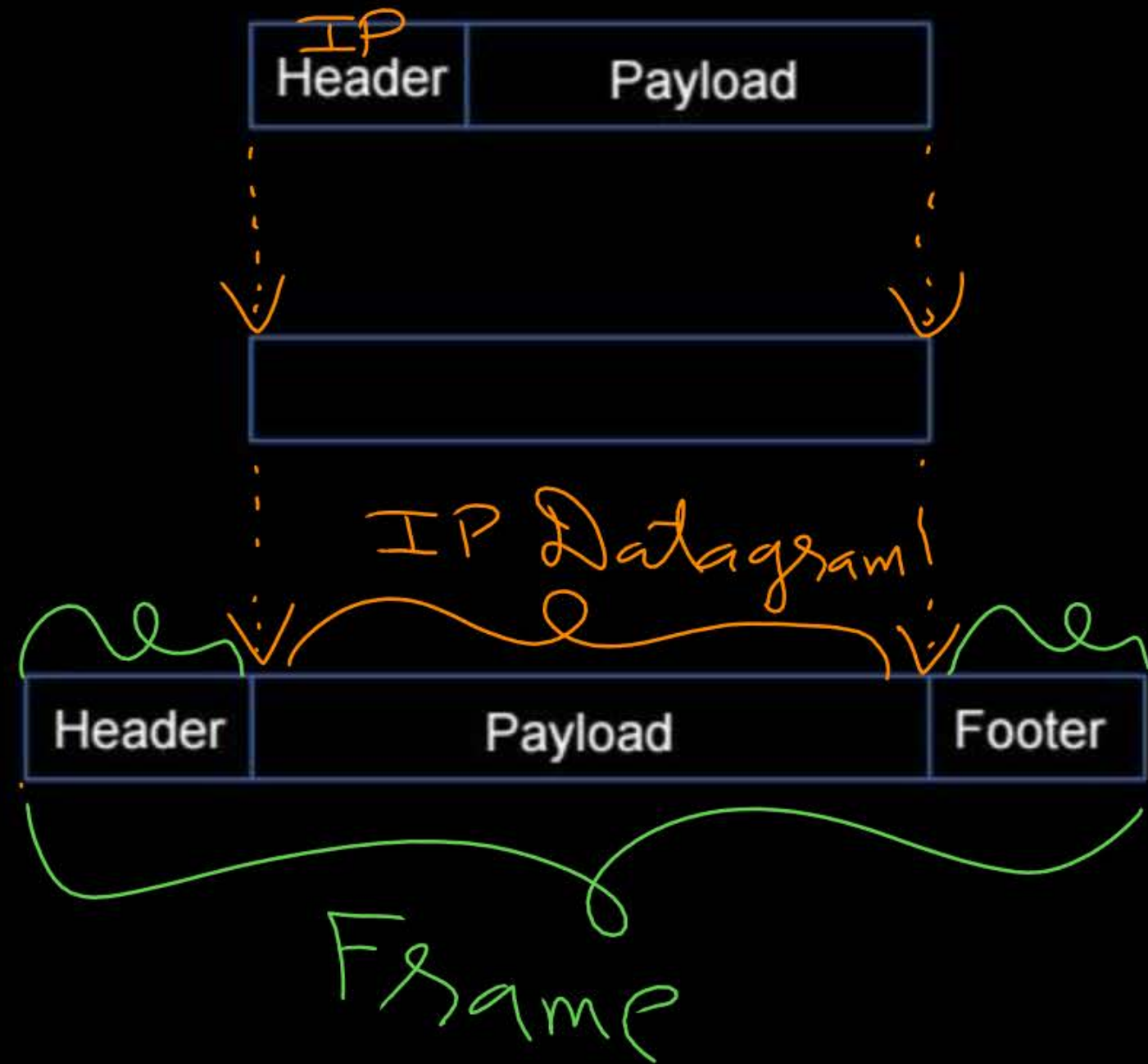
Framing

→ Receiver : Extract datagram from frame,
Datagrams passes to network layer

Network Layer PDU
IP "Datagram"

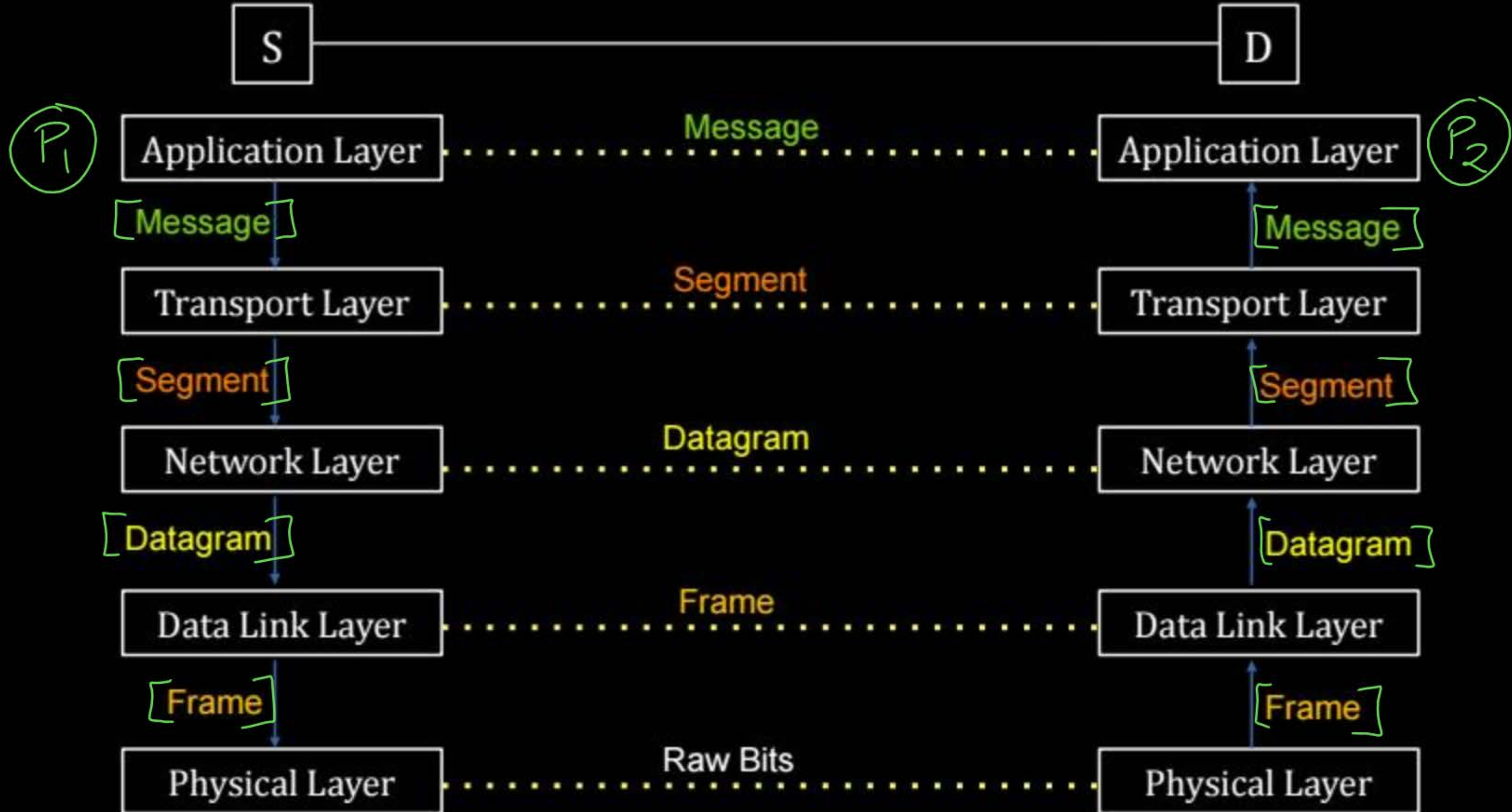
Data Link Layer SDU

Data Link Layer PDU
 "Frame"





Topic : Protocol Data Unit





Topic : Data Link Layer



-> Data Link Layer Networking Device : "Switch"

[Layer - 2 device, Old name : "Bridge"]

-> Store and Forward device

[Store, Process and Forward]

-> Forwarding based on MAC Address

[Forwarding/switching Table]



Topic : Physical Layer



=> Responsible for transmission of "bit"
[Bit synchronization between transmitter and receiver]

→ Copper Cable

→ Fiber Cable

→ Wireless

=> Encoding / Signaling → Transmit bit in the form of
Signal

[MCQ]

IIT-KGP

[GATE-2014][1 Mark]



#Q. In the following pairs of OSI protocol layer/sub-layer and its functionality, the **INCORRECT** pair is

- ☒ A Network layer and Routing *correct*
- ☒ B Data Link Layer and Bit synchronization *(Physical Layer) Incorrect*
- ☒ C Transport layer and End-to-end process communication *correct*
- ☒ D Medium Access Control sub-layer and Channel sharing *correct*

Ans: B

[MCQ]

IIT-D

[GATE-2004][1 Mark]



#Q. Choose the best matching between Group I and Group II

Group I	Group II
P. Data link layer	1. Ensures <u>reliable transport of data over a physical</u> [point-to-point link]
Q. Network layer	2. Encodes/decodes data for physical transmission $\Rightarrow$ physical layer
R. Transport layer	3. Allows <u>end-to-end communication between two processes</u>
	4. <u>Routes data from one network node to the next</u>

- ☒ A P - 1, Q - 4, R - 3
- ☐ B P - 2, Q - 4, R - 1
- ☐ C P - 2, Q - 3, R - 1
- ☐ D P - 1, Q - 3, R - 2

Ans: A



Topic : Physical Layer



→ Physical Layer Networking Device : “**Repeater** and **Hub**”
[Layer - 1 device]

→ **Hub** : Multi-port **Repeater**



Topic : Protocol Data Unit



Layer	PDU
Application	Message
Transport	Segment
Network	Datagram
Data Link	Frame

Physical → bit



Topic : Layer Services



Layer	Provide Services (to its upper layer)
Transport	Process-to-Process [end-to-end] Connectivity
Network	Host-to-Host Connectivity
Data Link	Node-to-Node Connectivity



Topic : Networking Devices



Layer	Networking Device
Application	Gateway
Network	Router
Data Link	Switch or [Bridge]
Physical	Hub, Repeater

(Layer - 7)
(Layer - 3)

→ Routing:
b/w different types
of networks
→ Protocol converter

→ Routing:
b/w similar type
of networks

[MCQ]

[IT-R]

[GATE-2025][1 Mark]



#Q. Identify the ONE CORRECT matching between the OSI layers and their corresponding functionalities as shown.

OSI Layers	Functionalities
(a) Network layer →	(I) Packet routing
(b) Transport layer ↗	(II) Framing and error handling
(c) Datalink layer ↗	(III) Host to host communication

~~A~~

(a)-(I), (b)-(II), (c)-(III)

B

(a)-(I), (b)-(III), (c)-(II)

~~C~~

(a)-(II), (b)-(I), (c)-(III)

~~D~~

(a)-(III), (b)-(II), (c)-(I)

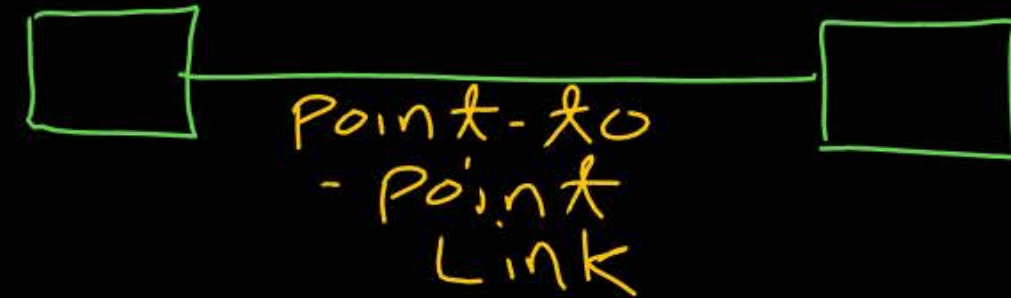
Ans: B



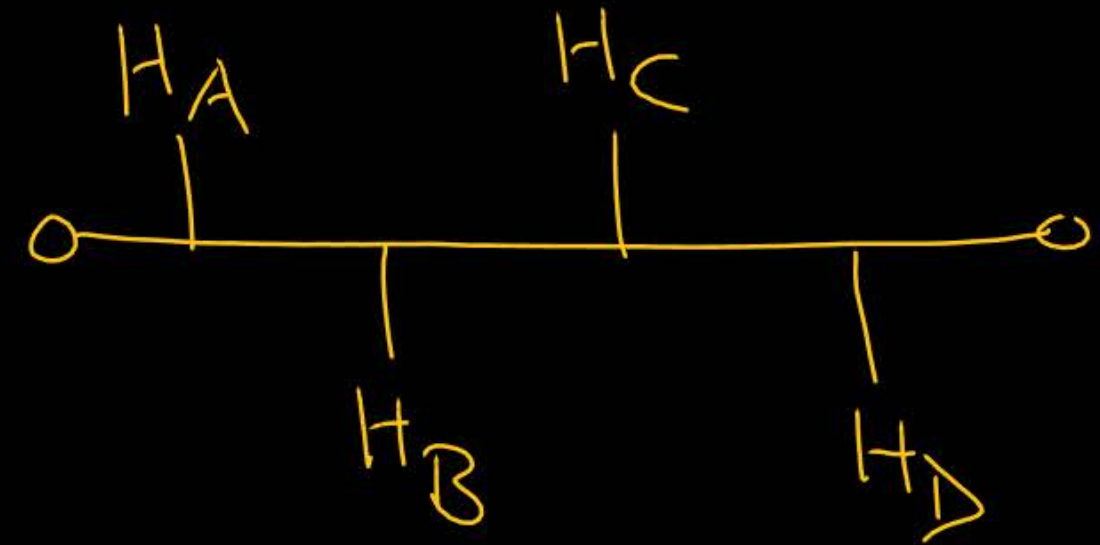
Topic : Line Configuration

→ Attachment of communication devices to a link.

1. Point-to-Point [PPP]
→ Dedicated link between two device
[One sender and one receiver]



2. Multipoint (Multidrop)
→ More than two devices share a single link
→ Broadcast medium
[One sender and all are receiver]
→ e.g. Bus topology, WiFi



[MCQ]

IIT-K

[GATE-2007][2 Mark]



#Q. Match the following :

List - I	List - II
P. SMTP	1. Application layer
Q. BGP	2. Transport layer
R. TCP	3. Data link layer
S. PPP	4. Network Layer
	5. Physical Layer

~~A~~ P - 2, Q - 1, R - 3, S - 5

B P - 1, Q - 4, R - 2, S - 3 ✓

~~C~~ P - 1, Q - 4, R - 2, S - 5

~~D~~ P - 2, Q - 4, R - 1, S - 3

Ans: B



Topic : Transmission Mode



→ Define the direction of signal flow between two linked devices.

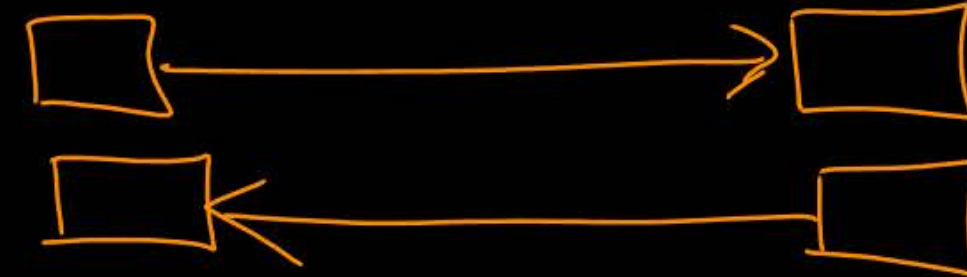
1. Simplex mode

→ One-way communication



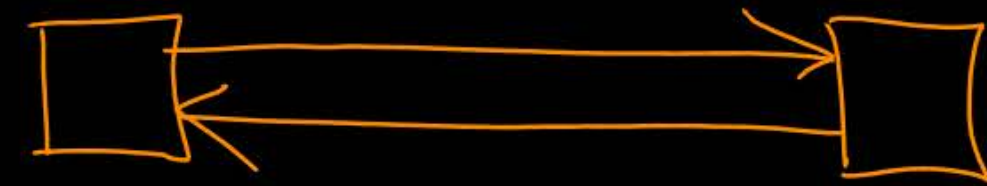
2. Half-Duplex mode

→ Either side communication at a time



3. Full-Duplex (Duplex) mode

→ Both side communication is possible at same time





Topic : Network Topology



- Arrangement of hosts inside a network.
- Different types of topology are :
 1. Mesh
 2. Star
 3. Bus
 4. Ring



Topic : Mesh Topology



→ [Fully Mesh]

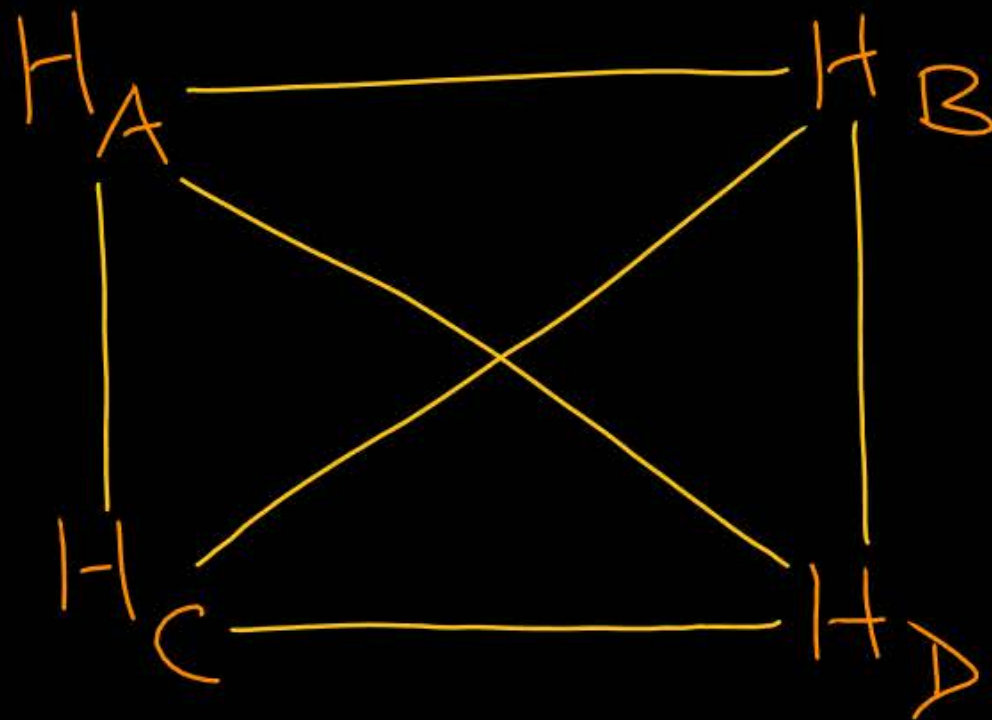
→ Every device has dedicated point-to-point link to every other device.

Total number of nodes = n ✓

Total number of links = $\boxed{{}^nC_2}$

$$= \frac{n * (n-1)}{2}$$

$$= O(n^2)$$





Topic : Mesh Topology



Advantages :-

- Faster Communication [in compare to all other topology]

Disadvantages :-

- Installation cost is very high
[Preferable for small area network]
- Inefficient utilization of links
- $(n - 1)$ input-output (I/O) port per device



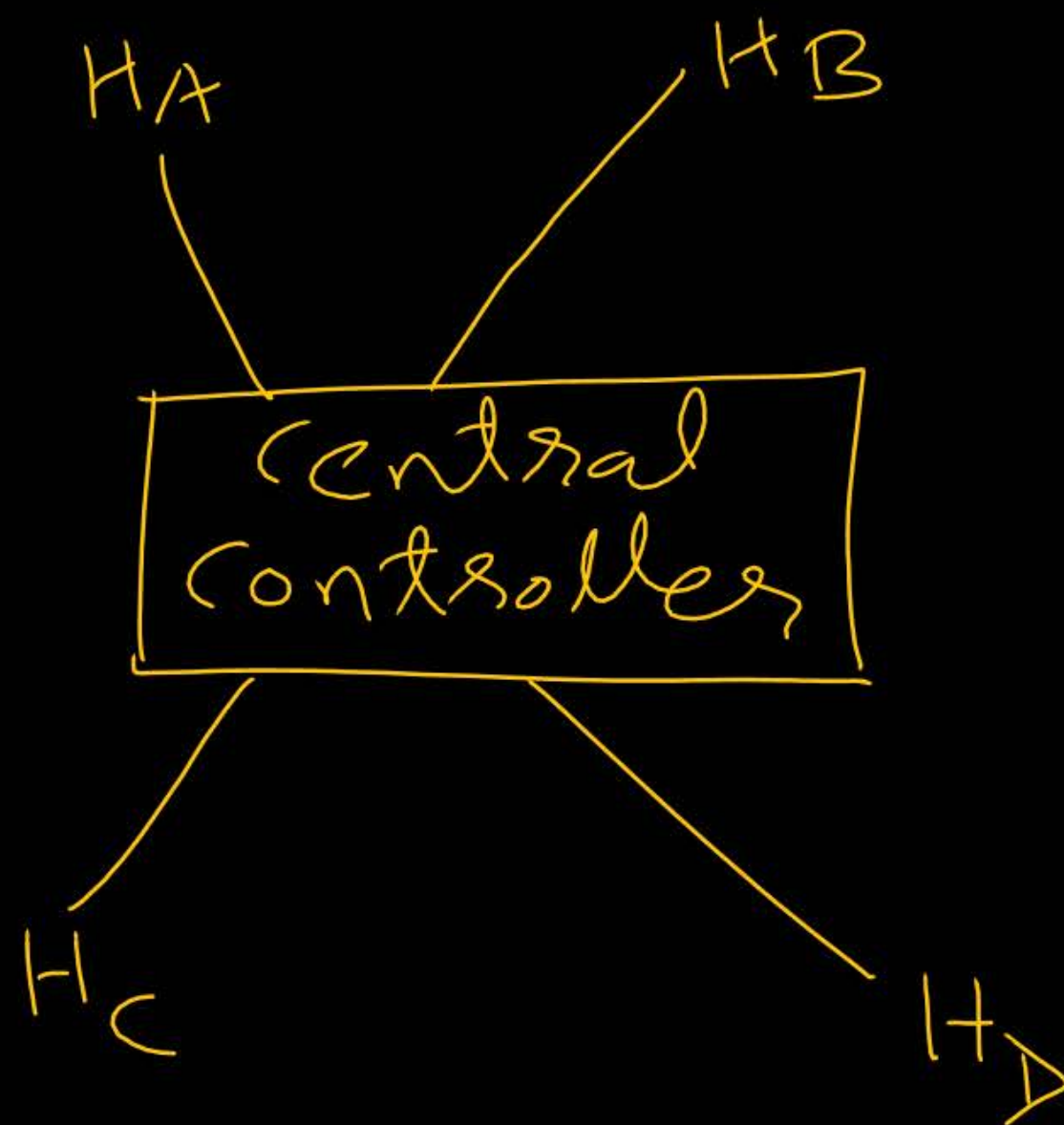
Topic : Star Topology

→ Every device has dedicated point-to-point link only to a central controller.

Total number of nodes = n

Total number of links = n

Central Controller : Hub, Switch or Router





Topic : Star Topology



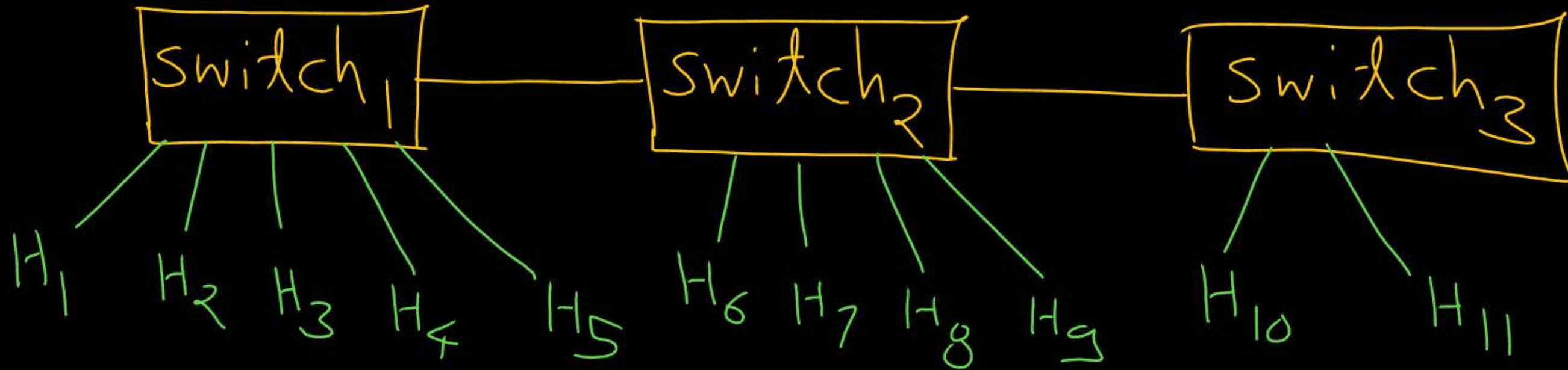
Advantages :-

- Insertion / removal of devices are easy
- Easy to extend the topology ✓

Disadvantages :-

- If central controller fails
then all communication stop

#Q. Consider that 11 machines need to be connected in a LAN using 6-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is ____.



Ans = 3



#Q. Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is ____.

[GATE 2019]

IIT-M, H.W.



2 mins Summary



Topic

Data Link Layer ✓

Topic

Physical Layer ✓

Topic

Network Topology

⇒ Switching
1) Circuit
2) Packet



THANK - YOU

